Presentation

What Drives Airline Profitability?

Operating-margin drivers, profitability classification and industry revenue forecast

Capstone Project | 30 global airlines, 2010-2026

By: NITHYA BHAVANE D

Context: Airline margins are thin and volatile

Problem Statement

Which operating and structural factors drive airline operating margin, can we classify profitable airline-years, and what is the outlook for industry revenue?

Objectives

- Describe and classify the data; test correlations
- OLS regression with feature selection (target: margin %)
- Logistic regression and decision trees (target: profitable?)
- Forecast revenue: MA, single and double exponential smoothing

30 airlines	**17** years (2010-26)
496 clean observations	**12** raw variables
7.2% avg margin ex-COVID	**-26.6%** avg margin 2020-21

Interpretation: Airline margins on average, it has been 7.2% per dollar sold in normal times. However, during the period of 2020-21 due to COVID, it was a loss of 26.6%. This one variable is so significant that it affects almost all the results of this project.

Target variables, feature variables and exclusions

Target variables

- operating_margin_pct (continuous) - OLS, regression tree
- profitable = margin > 0 (binary) - logistic, classification tree
- Industry revenue, $bn (time series) - MA, SES, Holt

Feature variables

- load_factor_pct
- log(revenue_usd_bn)
- passengers per aircraft (derived)
- covid flag (2020-21)
- business_model: legacy / low-cost / regional
- region: 5 regions

Excluded / identifiers

- operating_income_usd_bn: leakage
- revenue per passenger: VIF 12,375
- passengers, fleet size: redundant with revenue
- airline, IATA, country: identifiers
- year: time index only

- **Interpretation:** To select which factors we will be trying to predict (profit margin, whether an airline made a profit and total industry revenue) and factors that will be used as predictors. Operating income was not used as a factor because it is a product of revenue and profit margin, and its use would mean providing the model with the answer. Revenue per passenger was also not included in the model because its value does not vary much between airlines of the same kind (legacy, low cost or regional). Operating income was not used as a predictor because it is a product of revenue and profit margin. Its use would mean providing the model with the answer.
- Revenue per passenger was not used because its value does not vary much between airlines of the same kind. This overlap was measured using VIF, and the value of 12,375 means that the factor is problematic, because the acceptable values are below 10.
- Passenger and fleet sizes were omitted because their change closely correlates with the change in revenue.
- Airline name, IATA code and country are categorical variables and year is a time index.

Data quality checks and cleaning

Check	Result	Action
Missing values	**0**	**None needed**
Duplicate airline-years	**0**	**None needed**
Income = revenue x margin	**Max error $0.026bn**	**Rounding only**
Pre-launch row (Riyadh Air 2023)	**Fleet 0, passengers 0**	**Removed: 497 to 496 rows**
Revenue per passenger	**Legacy $667, low-cost $250, regional $333 (SD < 2)**	**Dropped from models (VIF 12,375)**

Interpretation: The data did not show any gaps or repeating rows and calculation mistakes, but only one mistake was identified in Riyadh Air 2023 was eliminated as it had 0 planes and 0 passengers as it had'nt started its operation yet. Also, it has been identified that revenue per passenger does not differ much in the airlines of the same type, hence causing statistical difficulties.

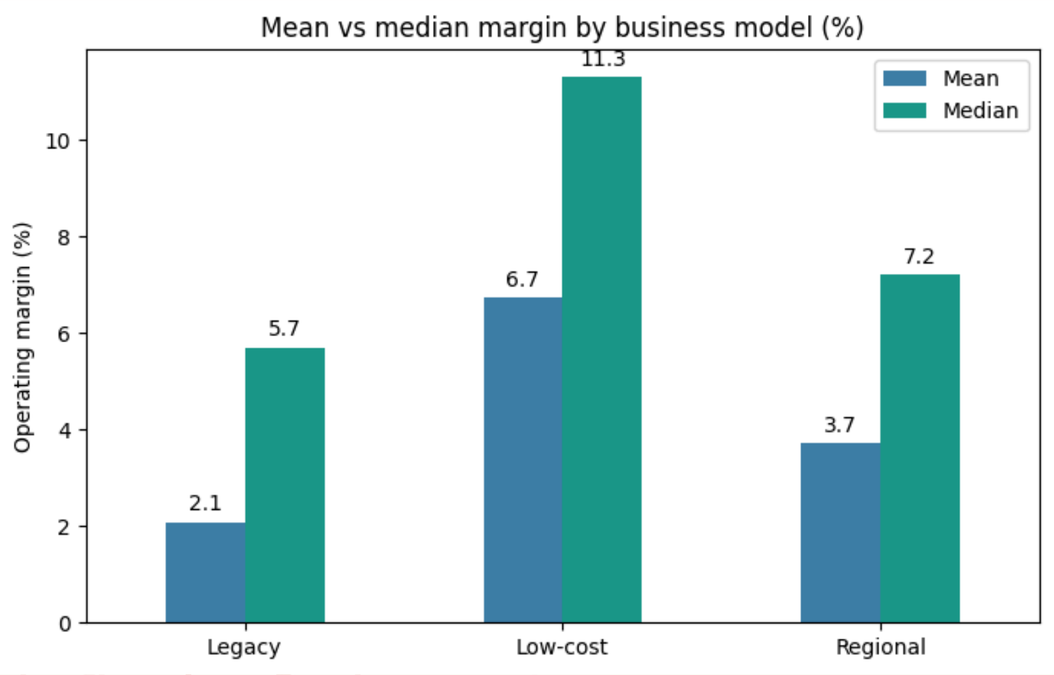
Descriptive statistics: central tendency and spread

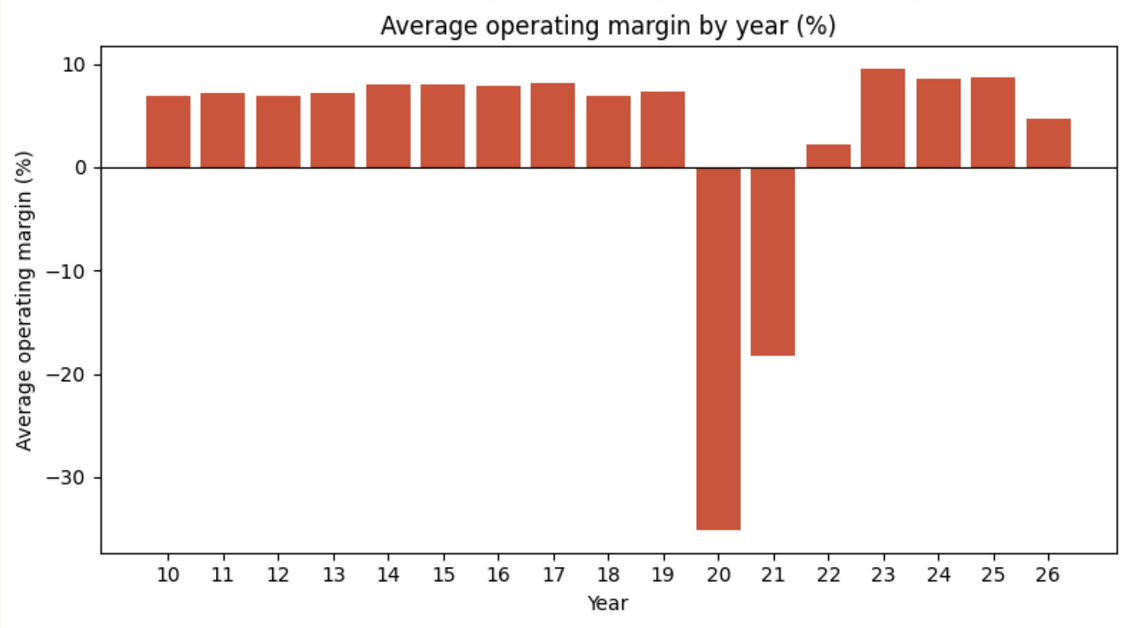
Variable	Mean	Median	Std dev	Min	Max	Skew
Revenue ($bn)	15.75	11.67	12.98	0.39	68.38	1.43
Operating margin	3.25	6.5	11.96	-44.1	17.7	-2.22
Operating income	0.73	0.73	1.54	-6.57	6.59	-0.8
Passengers (m)	28.38	20.9	21.67	0.8	129.3	1.49
Load factor (%)	81.5	83.6	7.93	50.7	92.8	-2.39
Fleet size	78.23	58	64.9	1	341	1.42

- The mean is the ordinary average. The median is the middle value when all records are lined up in order.
- The average margin (3.25%) is well below the median (6.5%). The COVID losses, which go as low as -44%, drag the average down, while the median is barely affected. So 6.5% is a fairer picture of a normal year.
- For revenue, passengers and fleet size, the average is higher than the median. A few giant airlines, with revenue up to $68bn and fleets up to 341 planes, pull the average up. The typical airline is smaller than the averages suggest.
- The skew column shows which direction the odd values lie. Positive skew means a few very large values. Negative skew (as in margin and load factor) means a few very low values.
- Load factor averages 81.5%, so on average about 4 out of 5 seats are filled.

Interpretation: The average margin (3.3%) is below the median (6.5%) due to the negative impact of the pandemic on the airlines' performance. Additionally, some of the biggest airlines contribute to the higher values of the averages concerning revenues, number of passengers, and aircraft. Thus, the typical company is smaller than it seems according to the averages.

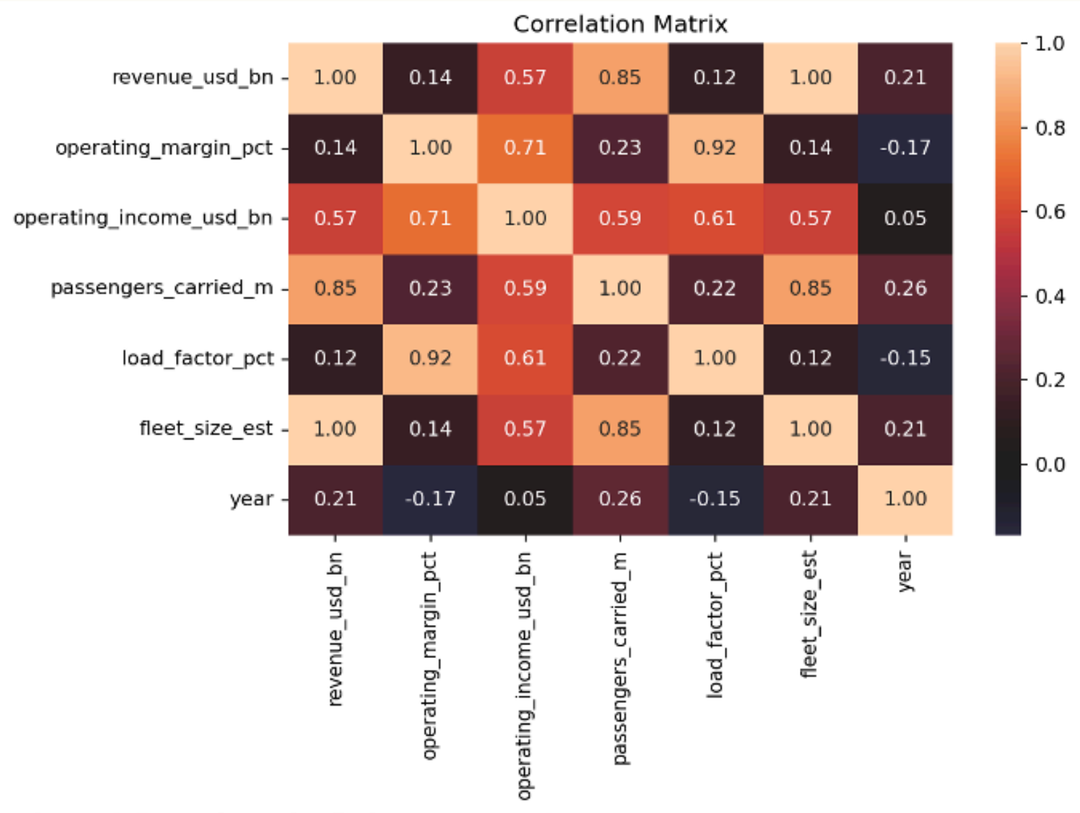
Margin by business model and by year

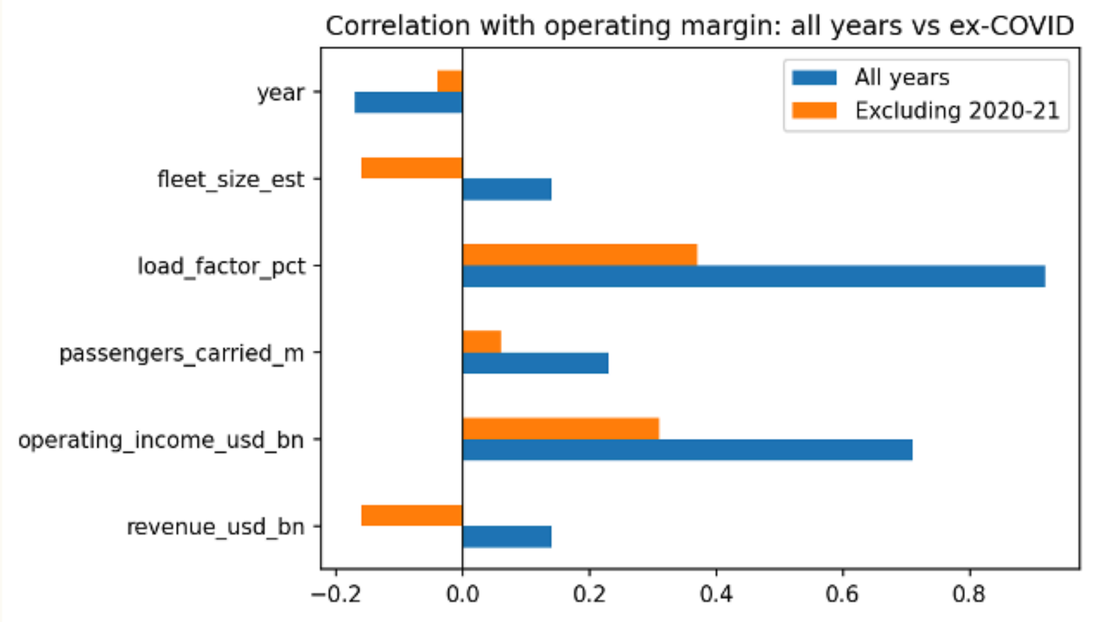




Interpretation: The left graph depicts the type of airlines on a vertical axis. Low-cost airlines perform the best with an average margin (median) of around 11%. Regional airlines follow them at around 7% margins, followed by legacy airlines (the traditional major airlines) at around 6% margins. The average for each category is less than their median because of the coronavirus losses. The graph on the right shows the margins over time. The margins have remained stable at around 7-8% from 2010 till 2019. However, they fell to -35% in 2020 and to around -18% in 2021. They bounced back from 2022 and were very strong in 2023. The fall in 2026 to 4.7% could actually be an indicator of something serious or the incompleteness of the 2026 data which needs verification.

Correlation with operating margin





Interpretation: The correlation coefficient is a measure ranging from -1 to +1 representing how closely two factors tend to move together. A value close to 0 represents lack of any relation while a value close to +1 represents a relationship where two things move up and down together. As per the heatmap below, the load factor (or the occupancy rate of planes) is 0.92 with a margin that seems to be the recipe for profitability. The graph on the right is crucial to understand the situation. Once 2020-21 is stripped out (orange bars), this correlation drops significantly to 0.37, which is way too weak. This is because during the coronavirus pandemic, planes were virtually empty and the airlines made losses together.

MLR (OLS): backward elimination leaves five significant drivers

Variable	Coef	Std err	p-value	p (HC3)	Std coef
Intercept	-59.47	5.07	<0.001	<0.001	-
Load factor (pct pt)	0.766	0.061	<0.001	<0.001	0.503
log(revenue)	0.635	0.268	0.018	0.062	0.048
COVID (2020-21)	-16.684	1.461	<0.001	<0.001	-0.447
Low-cost carrier	2.458	0.586	<0.001	<0.001	0.087
Latin America	1.688	0.829	0.042	0.036	0.034

0.895
train R2

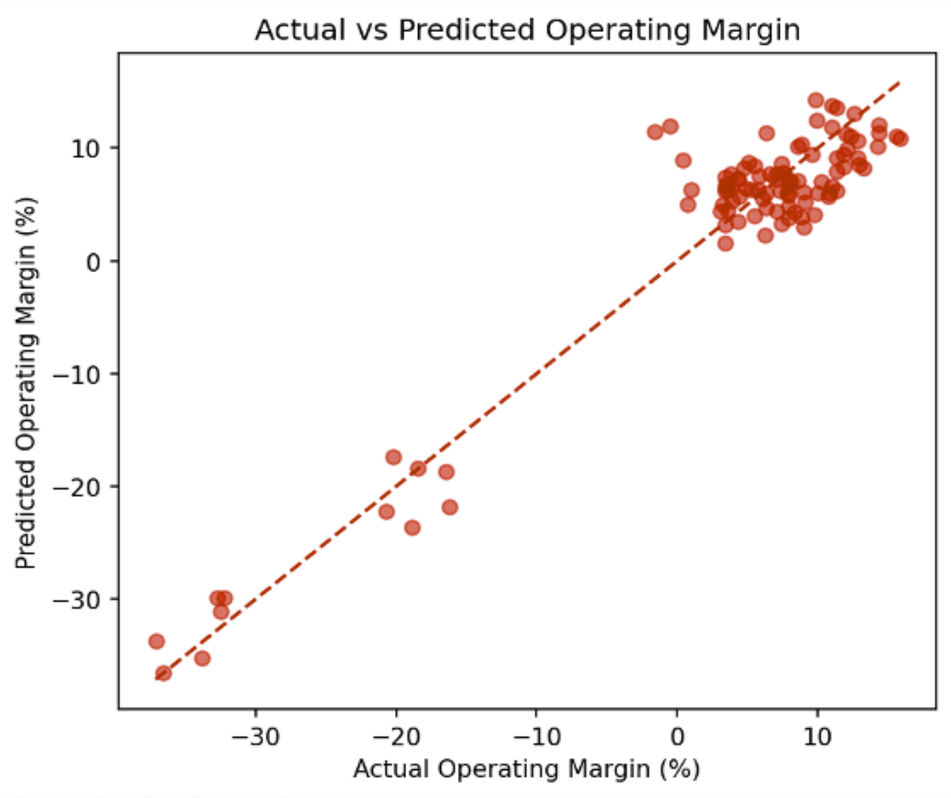
0.911
test R2

3.57
test RMSE

2.85
test MAE

Interpretation: This technique of multiple linear regression makes a formula that predicts the margin based on a number of different factors. The backward elimination starts with all the factors but eliminates the insignificant ones one by one until there are only five left:
COVID: a year of COVID reduces the margin by about 16.7 percentage points, the most significant coefficient.
Load factor: every additional percentage point of seats occupancy increases the margin by 0.77 percentage points.
Low-cost carrier: low-cost airlines have 2.5 more percentage points of margin than others.
Latin America: the airlines from Latin America have 1.7 percentage points more margin than others.
Size (log revenue): insignificant coefficient.
The R^2 of 0.895 on training data and 0.911 on test data show that the formula explains 90% of variations of the margin. "Test" data is data that was never used during learning, and the model performed just as well, so the result can be considered trustworthy. RMSE (3.57) and MAE (2.85) show that the prediction error is usually about 3 percentage points of margin.

OLS validation: fit is real, but COVID does most of the work



Validation findings

- Test R2 0.911 vs train 0.895: no overfitting
- Durbin-Watson 2.08: no autocorrelation
- Residuals non-normal (skew -1.79; Shapiro $p < 0.001$) because of COVID outliers; clustered SEs keep every driver significant
- Ex-COVID model (n = 438): only business model survives - low-cost +5.2 pts, regional +2.1 pts vs legacy; adj R2 only 0.28

OLS Results:

OLS Regression Results

Dep. Variable:	operating_margin_pct	R-squared:	0.895
Model:	OLS	Adj. R-squared:	0.894
Method:	Least Squares	F-statistic:	665.1
Date:	Fri, 02 Oct 2026	Prob (F-statistic):	2.22e-188
Time:	20:40:23	Log-Likelihood:	-1098.0
No. Observations:	396	AIC:	2208.
Df Residuals:	390	BIC:	2232.
Df Model:	5		
Covariance Type:	nonrobust		

Interpretation: Predictions by the model are highly correlated with actual values and there is no problem of overfitting (remembering rather than understanding the data). Nonetheless, without the years of the coronavirus outbreak, the model explains only 28% of variations among the airlines, and just the business model remains significant: low-cost airlines have 5 more points compared to legacy ones. To summarize, it is worth mentioning that the load factor is significant for the airline profitability because of its correlation with the COVID outbreak, but not due to any control of it.

Logistic regression: classifying profitable airline-years

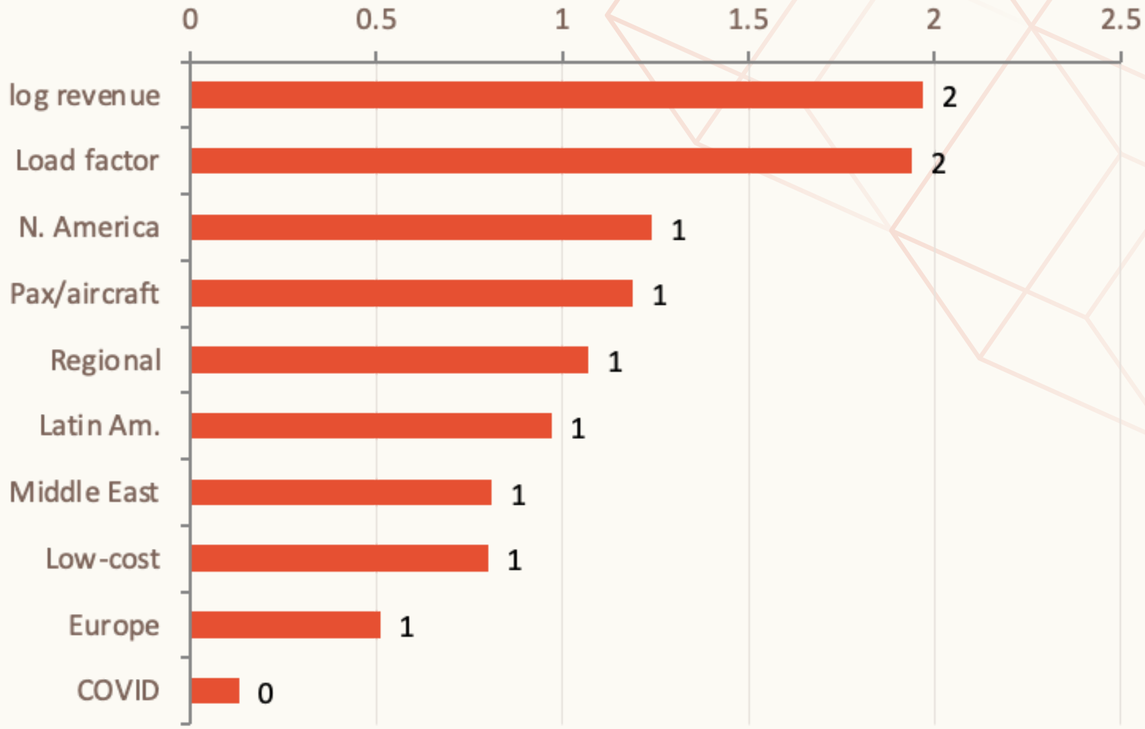
	Pred loss	Pred profit
Actual loss	**13**	1
Actual profit	0	**86**

0.990
accuracy (test)

0.935
AUC (test)

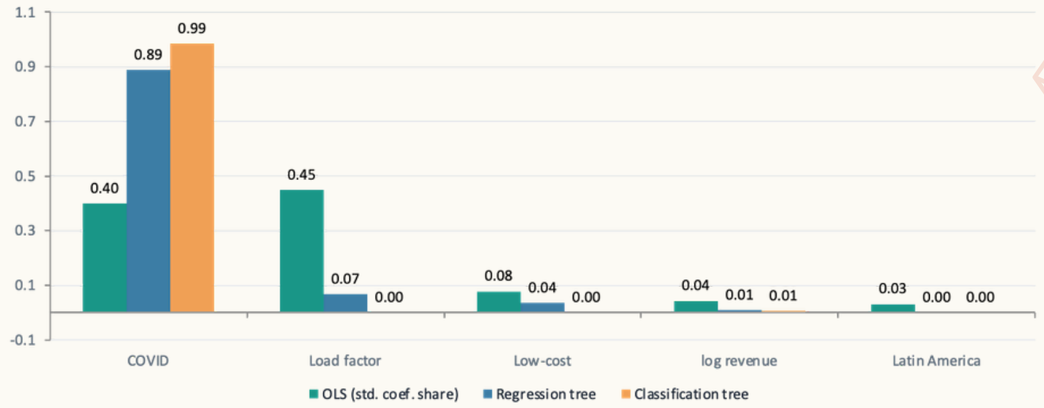
0.976
5-fold CV acc.

0.859
majority baseline



Interpretation: The confusion matrix is a scoreboard from 100 samples where 13 losses were identified correctly, 86 profits were identified correctly, and just one loss was falsely identified as profit. This gives us 99% accuracy. AUC is a score which tells us how well the model can rank risky years above safe years where 1.0 is perfection and 0.5 is random.
However, a basic rule, "assume profits unless the year is a COVID year" works at about 97.6% accuracy because none of the airlines had profits in 2020-21. Hence, the model doesn't provide much above the ability to identify COVID years. Similarly, the COVID coefficient shown in the bar chart is also not reliable. Mathematically, it is impossible to calculate the exact value of a coefficient which segregates two categories perfectly.

Linear vs tree-based feature importance



0.910

tree R2

0.990

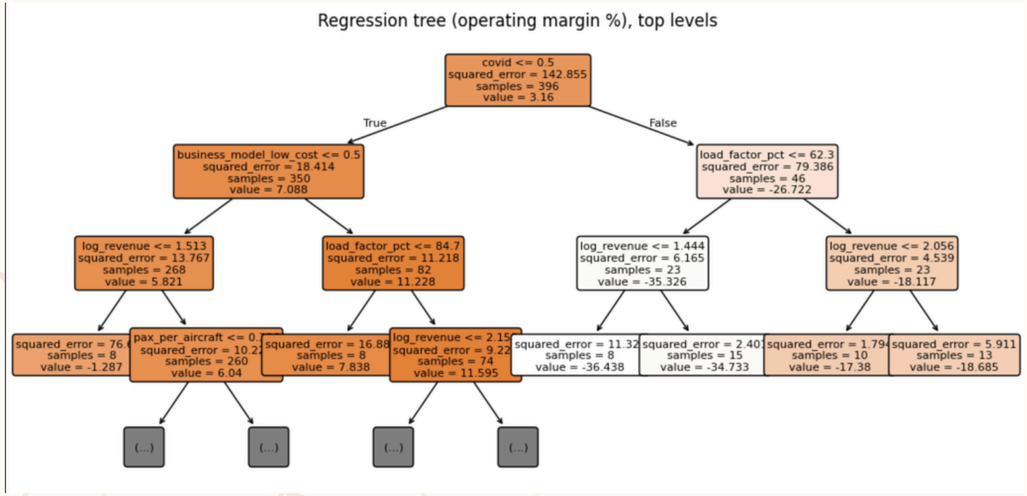
class. acc.

3.59

tree RMSE

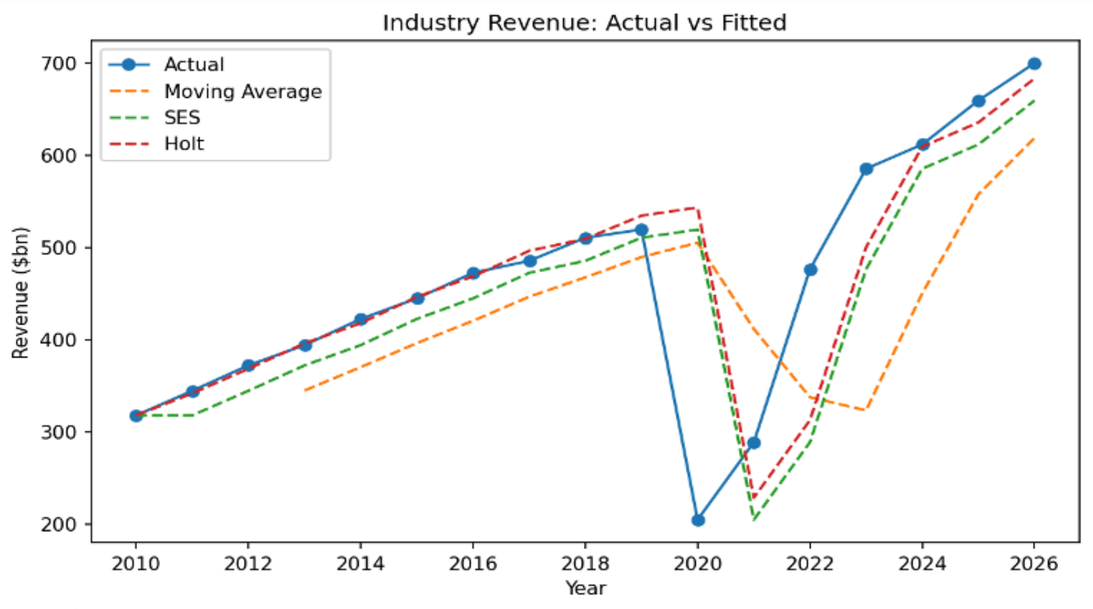
0.944

class. AUC



Interpretation: Decision trees use a chain of yes or no questions, and they are as effective as the regression techniques. However, they place 89-99% of the weight on the COVID flag, since it is already known whether it was a COVID year, and therefore the information about it provides much of the answers. Regression shares the responsibility between the COVID flag and the load factor, while decision trees place all of the weight on the first one.

Time series: moving average, single and double exponential smoothing



Metric	MA(3)	SES	Holt
All years			
MAE	105.9	68.4	52.1
RMSE	133.8	107.3	104.7
MAPE %	27.6	20.8	17.9
Ex-2020-22			
MAE	83.7	33.7	15.2
RMSE	107.1	42.6	27.8
MAPE %	15.2	6.2	2.6

Interpretation: Three techniques have been used to forecast the revenue of the industry, including MA (moving average technique for three years), SES (single exponential smoothing) and Holt method (double exponential smoothing). The most accurate method out of the above is Holt's. Normally, the prediction made is accurate by an error of only about 2.6% but the error rate during the period 2020-2022 is very high due to the effect of the pandemic which no one can predict. Both SES and Holt's give full weight to the last year's figure and this makes the predictions react to current events.

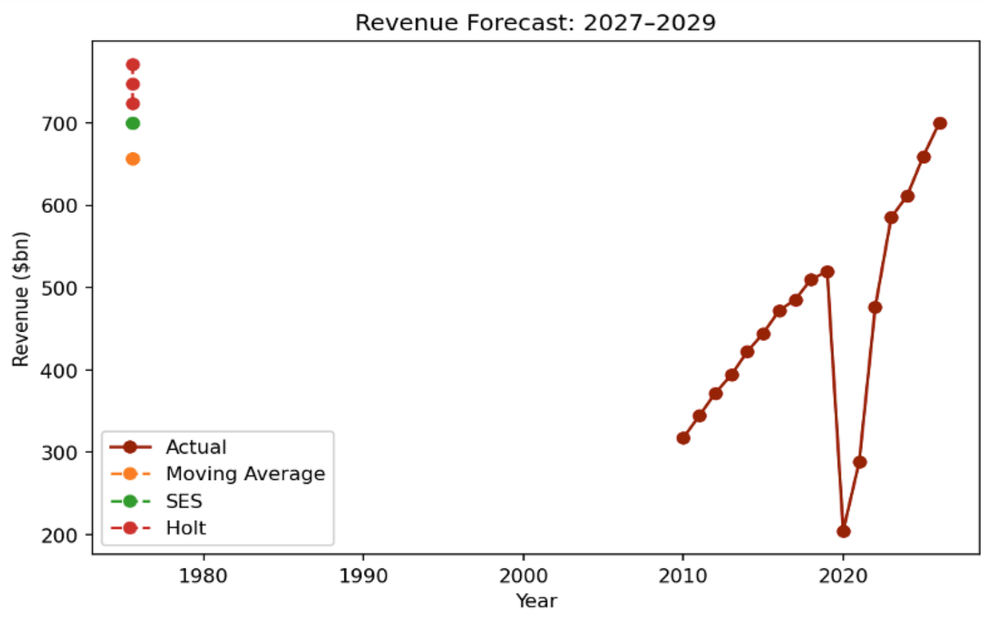
This method forecasts the total revenue of the industry. Three methods have been used in this forecasting process:

MA3 - Moving Average

SES - Single Exponential Smoothing

Holt's Double Exponential Smoothing

Time series: moving average, single and double exponential smoothing



Year	MA(3)	SES	Holt
2027	656.8	699.7	**723.6**
2028	656.8	699.7	**747.5**
2029	656.8	699.7	**771.3**

Interpretation: The growth forecast for Holt is consistent at about $24 billion annually, hitting a total of about $771 billion in 2029. The three-year smoothed forecast ($657 billion) is too conservative as it still includes earlier years. In addition, the simple smoothing forecast does not have any growth potential because it only uses the $700 billion figure from last year.

Conclusion

- Across 30 airlines from 2010 to 2026, the 2020–21 COVID shock is the dominant driver of operating margin. Margins averaged 7.2% in normal years but -26.6% during the shock, and no airline was profitable in either of those years.
- The OLS model (test R^2 0.911) and both decision trees agree.
- COVID alone costs about 16.7 margin points, and load factor looks like a strong driver (r = 0.92) only because it collapsed at the same time.
- Excluding 2020–21, that correlation falls to 0.37, and the only significant structural driver is business model, with low-cost carriers earning about 5 points more than legacy airlines (adjusted R^2 0.28).
- Profitability classification reaches 0.99 accuracy, but simply predicting "profitable unless COVID" scores 0.976, so the model adds little beyond identifying shock years.
- For revenue, Holt's double exponential smoothing fits best (MAPE 2.6% in calm years) and projects industry revenue rising from about $724bn in 2027 to $771bn in 2029, roughly $24bn a year.
- This is only indicative, since the 2026 base year may be partial and the trend assumes post-COVID growth continues. For airlines, the practical implication is that resilience to external shocks and a low-cost model matter more for profitability than load factor or size alone.

Code and outputs (GitHub / Kaggle): https://github.com/NithyaBhavaneD , https://www.kaggle.com/nithyabhavanerajan